Example 3.23

If the centre of the circle $x^2+y^2+2x-6y+1=0$ lies on a straight line $ax+2y+2=0$, then find the value of 'a'

Solution

Centre C(–1, 3)

It lies on $ax+2y+2=0$

$$-a+6+2=0$$

$$a=8$$

Example 3.24

Show that the point (7, –5) lies on the circle $x^2+y^2-6x+4y-12=0$ and find the coordinates of the other end of the diameter through this point.

Solution

Let $A(7,-5)$

Equation of circle is

$$x^2+y^2-6x+4y-12=0$$

Substitute (7, –5) for (x, y) , we get

$$x^2+y^2-6x+4y-12 =$$

$$7^2+(-5)^2-6(7)+4(-5)-12$$

$$= 49 + 25 - 42 - 20 - 12 = 0$$

$\therefore$ (7, –5) lies on the circle

Here $g = -3$ and $f = 2$

$\therefore$ Centre = $C(3,-2)$

Let the other end of the diameter be $B(x, y)$

Midpoint of AB = $\left(\frac{x+7}{2},\frac{y-5}{2}\right)=C(3,-2)$

$\frac{x+7}{2}=3$ | $\frac{y-5}{2}=-2$

$x=-1$ | $y=1$

Other end of the diameter is (–1 , 1).

Example 3.25

Find the equation of the circle passing through the points (0,0), (1, 2) and (2,0).

Solution

Let the equation of the circle be $x^2+y^2+2gx+2fy+c=0$

The circle passes through the point (0, 0)

$$c=0 \qquad \ldots (1)$$

The circle passes through the point (1, 2)

$$1^2+2^2+2g(1)+2f(2)+c=0$$

$$2g+4f+c=-5 \qquad \ldots (2)$$

The circle passes through the point (2, 0)

$$2^2+0+2g(2)+2f(0)+c=0$$

$$4g+c=-4 \qquad \ldots (3)$$

Solving (1), (2) and (3), we get

$g=-1$, $f=-\frac{3}{4}$ and $c=0$

$\therefore$ The equation of the circle is

$$x^2+y^2+2(-1)x+2\left(\frac{-3}{4}\right)y+0=0$$

ie., $2x^2+2y^2-4x-3y=0$

3.4.4 Parametric form of a circle

Consider a circle with radius r and centre at the origin. Let $P(x, y)$ be any point on the circle. Assume that OP makes an angle θ with the positive direction of x-axis.

Draw PM perpendicular to x-axis.

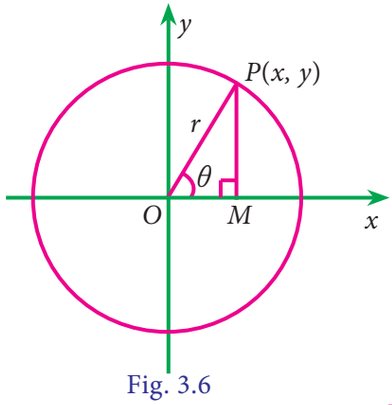


Fig. 3.6

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